ISTA 130 Strings I Worksheet

What is the output from the following lines of code?

```
text = 'Rocket!'

print(text[2])

print(text[-1] * 5)

mystery = text[-2] + text[0] + text[-3] + text[4]

m2 = mystery.upper()

print(m2)

print(mystery)

print(m2 + m2.lower())

print(m2 > m2.lower())

print(m2 < m2.lower())

print('REE' in m2)

print('REE' in m2.lower())

for i in range(len(text)):  # what kind of traversal is this?
     if i % 2 == 1:
          print(text[i])


for ch in text:  # what kind of traversal is this?
     if ch > 'e':
          print(ch)

def odds(string):
     new_str = ''
     for i in range(len(string)):
          if i % 2 == 1:
               new_str += string[i]
     return new_str

print(odds(text))
```

Write a function called `first_and_last` that takes one string argument and returns a new string containing the first and last characters in the argument. If the argument is the empty string return `None`.

Write a function called `thirds` that takes one string argument and returns a new string consisting of the third, sixth, ninth, etc. characters in the argument. If there aren't at least three characters, return `None`. What kind of traversal did you use?

Write a function called `count` that takes two string arguments. The first is a nonempty string and the second is a single character. Return the number of times the character appears in the string. What kind of traversal did you use?

Write a function called `smallest` that takes one nonempty string argument and returns the smallest character that is not a space, tab, or newline. Assume the first character in the string is not a space, tab, or newline. What kind of traversal did you use?

Write a function called `alphabetize` that takes three string arguments and prints them in alphabetical order, one per line, starting with the smallest. Do not distinguish between capital and small letters.